

# ARCHANA J

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## PROFILE

Motivated B.Tech student in Artificial Intelligence and Data Science with a strong focus on designing efficient, scalable solutions and translating data into actionable insights. Demonstrates expertise in problem decomposition, performance optimization, and developing real-world applications. Adept at bridging analytical thinking with practical implementation in dynamic environments. Committed to continuous learning, innovation, and delivering high-impact solutions in fast-paced, collaborative settings.

## SKILLS

- **Languages:** Python, SQL, Java, C/C++, R
- **AI / ML:** Machine Learning, RAG, LLM APIs, Random Forest, Decision Tree, Naïve Bayes, Scikit-learn, Pandas, NumPy, TensorFlow, Neural Networks, Deep Learning, Data Preprocessing, Data Analysis, Feature Engineering, LangChain, Google Gemini
- **DevOps & Tools:** Git, GitHub, Docker, Postman, REST APIs, VS Code, CI/CD, Figma, IoT (ESP32), Make.com
- **Databases:** MongoDB, SQL
- **Data Analytics:** Power BI, Microsoft Excel
- **Frameworks & Libraries:** Flask, FastAPI, Streamlit, React Native
- **Cloud & Deployment:** AWS, GCP, API Deployment, Containerization, Cloud Fundamentals
- **Core Concepts:** Data Structures & Algorithms, OOP, DBMS, Operating Systems

## WORK EXPERIENCE

The Hindu

MACHINE LEARNING INTERNSHIP

AUG 2025 - OCT 2025

- Developed a Machine Learning based - Fake News Detection System, achieving **94% classification accuracy** across a dataset of **1,000+ news articles** to distinguish real vs. misleading content
- Built an Email Chatbot to automate responses, **reducing manual reply time by 70%** and handling 500+ queries with **89% intent recognition accuracy**
- Designed an interactive English Language Learning Game with 15+ step-based progression levels, **improving learner engagement by 60%** and supporting 200+ vocabulary exercises

## PROJECTS

STUDENT ATTENDANCE MONITORING AND REMINDER SYSTEM FOR EXAMS | Ed Tech | GITHUB

- Developed a predictive analytics system using Naive Bayes and Decision Tree with data preprocessing, feature engineering, and model evaluation on **3,000+** student records, achieving **92% accuracy** and reducing misclassification by **18%**.
- Designed an end-to-end attendance tracking and eligibility prediction system with Make.com automation, reducing manual follow-ups by **85%** and improving communication turnaround by **60%**. Enabled proactive identification of at-risk students.
- **Tech Stack:** Python, Scikit-learn, R, Make.com

CROP YIELD PREDICTION SYSTEM | AgriTech | GITHUB

- Implemented Crop Yield Prediction System using Random Forest algorithm trained on **5,000+** historical records, **achieving 91% prediction accuracy** to forecast optimal crop selection based on location and season
- Predicted crop yield quantities and market pricing with **87% accuracy**, enabling farmers to make data-driven decisions and improve agricultural planning efficiency by 40%
- **Tech Stack:** Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Jupyter Notebook, CSV/Excel

## EDUCATION

B.Tech Artificial Intelligence and Data Science | CGPA: 9.49/10

2023 - MAY 2027

Saveetha Institute of Medical and Technical Sciences

## CERTIFICATES

- Introduction to Internet of Things - NPTEL | NPTEL25CS44S243308626 | April 2025
- Privacy and Security in online Social Media - NPTEL | NPTEL25CS117S675200190 | October 2025
- Master in Data Science - UDEMY | UC-139bfd91-aa7c-4a70-94c1-8ad97529d500 | July 2025
- SQL Query & Manage Database - UDEMY | UC-ea17e013-9b99-4cab-bc61-bc9b0dcdda23 | July 2025